

Research Compendium Structures Using rrtools

Based on Marwick, Boettiger & Mullen (2018)

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Introduction

This document presents the recommended file structures for research compendia as described by Marwick, Boettiger, and Mullen (2018) in “Packaging Data Analytical Work Reproducibly Using R (and Friends)” published in *The American Statistician* (72:1, 80-88). The `rrtools` package was developed to bootstrap creation of these structures.

A research compendium follows three core principles: 1. Organize files according to community conventions 2. Maintain clear separation of data, method, and output 3. Specify the computational environment

Figure 1: Small Research Compendium

The simplest compendium structure contains only essential elements: metadata, documentation, data, and analysis code.

```
COMPENDIUM
|
├─ DESCRIPTION ..... project metadata & dependencies
├─ README.md ..... description of contents and guide to users
├─ data/
│   └─ my_data.csv .... raw data in open formats, not changed once created
└─ analysis/
    └─ my_script.R .... R code used to analyse and visualise data
```

Caption: An example of a small research compendium. This minimal structure fulfills the three generic principles: it organizes files according to convention, separates data and code, and specifies the computational environment through the `DESCRIPTION` file. Real-world example: Duffy (2015) parasite study archived at [doi:10.5281/zenodo.17804](https://doi.org/10.5281/zenodo.17804).

Figure 2: Small Compendium with Dynamic Document

This intermediate structure adds a `LICENSE` file and replaces the R script with an R Markdown dynamic document that interweaves code and narrative text.

COMPENDIUM

```
|
|— DESCRIPTION ..... project metadata & dependencies
|— README.md ..... description of contents and guide to users
|— LICENSE ..... specify conditions of use/reuse of code, data,
|                  text and output
|— data/
|   └─ my_data.csv .... raw data in open formats, not changed once created
└─ analysis/
    └─ my_report.Rmd .. R Markdown file with R code and text interwoven
```

Caption: *An example of a small research compendium that also includes a license file and a dynamic document. The R Markdown file generates all data visualizations and statistical content of the publication with commentary on methods. Real-world example: Conrad et al. (2016) zooarchaeological study archived at University of New Mexico.*

Figure 3: Medium Research Compendium

A more complete use of the R package structure, adding custom functions in `R/` with auto-generated documentation in `man/`, plus a `NAMESPACE` file for function exports.

```
COMPENDIUM
|
|— DESCRIPTION ..... project metadata & dependencies
|— README.md ..... description of contents and guide to users
|— LICENSE ..... specify conditions of use/reuse of code, data,
|                  text and output
|— NAMESPACE ..... auto-generated file exporting R functions for
|                  repeated use
|— data/
|   └─ my_data.csv .... raw data in open formats, not changed once created
|— analysis/
|   └─ my_report.Rmd .. R Markdown file with R code and text interwoven
|— R/
|   └─ my_functions.R . custom R functions used repeatedly throughout
|                  the project
|— man/
|   └─ my_functions.Rd auto-generated documentation for the custom R
|                  functions
```

Caption: *An example of a more complete use of the R package structure for a research compendium. The `R/` directory contains custom functions used throughout the project. The `man/` directory contains auto-generated documentation (via `roxygen2`). The `NAMESPACE` file exports functions for use in the current project and by other packages. Real-world example: Hollister et al. (2016) lake trophic modeling study archived at [doi:10.5281/zenodo.40271](https://doi.org/10.5281/zenodo.40271).*

Figure 4: Large Research Compendium

The most elaborate structure incorporates software engineering tools: continuous integration, containerization, build automation, and unit testing.

```

COMPENDIUM
|
|— DESCRIPTION ..... project metadata & dependencies
|— README.md ..... description of contents and guide to users
|— LICENSE ..... specify conditions of use/reuse of code, data,
|                  text and output
|— NAMESPACE ..... auto-generated file exporting R functions for
|                  repeated use
|— .travis.yml ..... continuous integration service hook for auto-testing
|                  at each commit & push to GitHub
|— Dockerfile ..... makes a custom isolated computational environment
|                  for the project
|— analysis/
|   |— data/
|   |   |— my_data.csv raw data in open formats, not changed once created
|   |— scripts/ ..... code files (R, Shell, etc.) for data cleaning,
|   |                  analysis, visualisation
|   |— my_report.Rmd .. R Markdown file with R code and text interwoven
|   |— Makefile ..... builds a PDF/DOCX/HTML file from the Rmd, code
|   |                  & data
|— R/
|   |— my_functions.R . custom R functions used repeatedly throughout
|   |                  the project
|— man/
|   |— my_functions.Rd auto-generated documentation for the custom R
|   |                  functions
|— tests/
|   |— testthat.R ..... unit tests of R functions to ensure they perform
|   |                  as expected

```

Caption: An example of a more elaborate research compendium based on the R package structure. This structure includes: (1) `.travis.yml` for continuous integration that automatically tests the package build after each Git commit; (2) `Dockerfile` for containerization that preserves exact versions of R and all dependencies; (3) `Makefile` for build automation that specifies relationships between inputs, outputs, and code; (4) `tests/` directory for unit testing custom functions. Real-world example: Boettiger et al. (2015) fisheries management study archived at [doi:10.5281/zenodo.12669](https://doi.org/10.5281/zenodo.12669).

Summary Comparison

Component	Small	Small+Rmd	Medium	Large
DESCRIPTION	X	X	X	X
README.md	X	X	X	X
LICENSE		X	X	X
NAMESPACE			X	X
data/	X	X	X	X
analysis/	X	X	X	X
R/			X	X
man/			X	X
tests/				X

Component	Small	Small+Rmd	Medium	Large
Makefile				X
Dockerfile				X
.travis.yml				X

References

Marwick B, Boettiger C, Mullen L (2018). “Packaging Data Analytical Work Reproducibly Using R (and Friends).” *The American Statistician*, 72:1, 80-88. DOI: 10.1080/00031305.2017.1375986

rrtools package: <https://github.com/benmarwick/rrtools>